ordered pair

Two numbers (called *x*-coordinate and *y*-coordinate) written in a certain order.

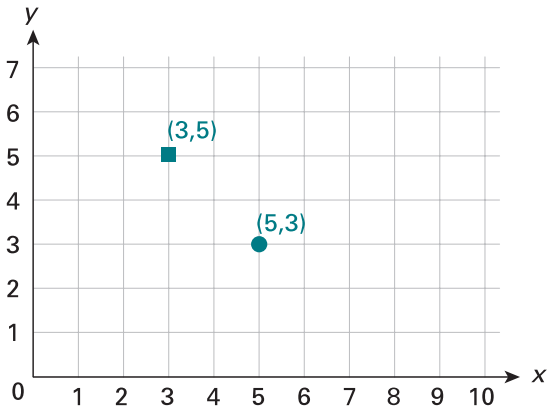
Ordered pairs are usually written between brackets.

Example

(5, 3)

The *x*-coordinate is always written first.

The ordered pair □ (3, 5) is not the same as the ordered pair ○ (5, 3).



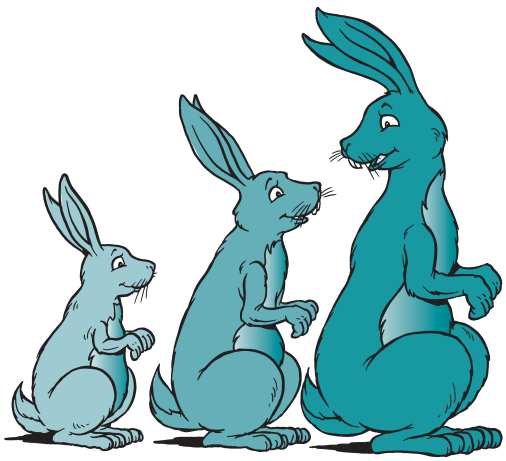
See axis, brackets, coordinates

ordering

Placement according to size, colour, numerical value, etc.

Example

The rabbits have been ordered from smallest to largest.



order of operations

Used when evaluating complex number sentences.

(i) Number sentences with grouping symbols.

When grouping symbols are used, working is done from inside the brackets out.

Example

$5\{3 - [(4 \times 9) - (20 - 4)] + 19\}$

$= 5\{3 - [36 - 16] + 19\}$

$= 5\{3 - 20 + 19\}$

$= 5 \times 2$

$= 10$

(ii) When no grouping symbols are used, starting from the left do all multiplications and divisions, then again from the left, do all additions and subtractions.

Example

$48 \div 3 + 2 - 4 \times 3$

$(48 \div 3) + 2 - (4 \times 3)$ **Insert brackets around multiplication and division.**

$= 16 + 2 - 12$ **Do addition first, then subtraction.**

$= 18 - 12$

$= 6$

(iii) Sometimes 'of' is used.

Example

$5\ (3 + 8) - \frac{1}{2}$ of 10

$= 5 \times 11 - (\frac{1}{2} \times 10)$

$= 55 - 5$

$= 50$

Note: To remember the order of operations, do: Brackets first, Of, Division, Multiplication, Addition, Subtraction.

Think : BODMAS.

See braces, brackets, operation, parentheses